

Richie Tan

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science, GPA: 3.73/4.0

August 2023 – May 2027

Relevant Coursework: Data Structures and Algorithms, Analysis of Algorithms, Systems Programming, Operating Systems, Cloud Computing, Relational Databases, Intro to Artificial Intelligence, Data Mining and Machine Learning

EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

Capital One

Richmond, VA

- Built Connex Autofix, a Claude tool that turns Jira tickets into reviewable Lit/TypeScript PRs
- Saved 360+ developer hours and \$21,900 across a 116-ticket backlog, with recurring savings on new tickets
- Drove design-team adoption with live demos, hands-on install support, and fast issue turnaround
- Reduced component review from 10 minutes to under 30 seconds with a before-and-after render pipeline

Founding Software Engineer

August 2024 – Present

AlphaSeeker AI

Boston, MA (Remote)

- Built a Next.js financial insights platform end-to-end, from initial architecture to production deployment
- Implemented an automated error-tracking system with email notifications and detailed, traceable logging
- Automated daily price forecasting for 26,000 stock tickers with nearly 60% directional accuracy
- Reduced search load times from 20 seconds to near-instant by optimizing PostgreSQL queries and adding indexes
- Led a team of 3 developers to build forecast visualization interfaces, enabling price prediction comparisons
- Developed an agentic AI financial analyst that provides data-driven market insights through voice and text

Software Engineering Intern

May 2025 – August 2025

Pacific Northwest National Laboratory

Richland, WA

- Reduced SQL query latency from 10 seconds to sub-second by replacing JOINS with an indexed cache table
- Built a RAG-powered Semantic Kernel chat agent in C# enabling natural-language access to internal APIs
- Improved LLM response accuracy using semantic search and text embeddings, reducing hallucinations
- Simplified the contacts database API and optimized database schema to support the new chat agent backend

Scientific Applications Assistant

May 2024 – Present

Purdue's Rosen Center for Advanced Computing (RCAC)

West Lafayette, IN

- Shipped a redesigned web dashboard across 4 HPC clusters, serving 12,000+ users
- Reduced load times by 50% by implementing Ruby on Rails backend caching and IndexedDB frontend caching
- Fixed accessibility issues across the dashboard, confirmed by center staff and HPC user testing
- Open-sourced the dashboard and presented a co-authored workshop paper at Supercomputing 2025

PROJECTS

Personal Website | *Next.js, TypeScript, Claude API*

- Recreated the Claude Code CLI's exact UI, animations, and behavior in the browser from scratch
- Wired it to the Claude API with tool calls that load my resume and GitHub stats to answer questions

PDFNotes | *Next.js, TypeScript, Google Gemini, react-pdf*

- Built a Gemini-powered PDF reader with streaming, multi-turn chat for analyzing documents
- Implemented text/screenshot selection, LaTeX math rendering, and tabbed multi-document support

PrePrompt | *TypeScript, Node.js, Turborepo*

- Published an open-source CLI tool on npm that benchmarks prompts on AI coding agents in parallel sandboxes
- Automated verifiable success criteria generation from prompts and agent output evaluation

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C, SQL, Bash

Frameworks/Databases: Next.js, FastAPI, Ruby on Rails, Tailwind CSS, PostgreSQL, Redis

Developer Tools: Git, GitHub Actions, Docker, GCP, Linux, Vitest, pytest

Libraries: React, pandas, NumPy, matplotlib